

## **What was the main issue identified?**

### **Refactoring for Single Responsibility (SRP)**

The original `addTask()` method was overloaded with multiple responsibilities, including task validation, queue population, notification handling, logging, and queue execution. To establish a clearer separation of concerns, I decoupled the logging and notification mechanisms from the core queue management logic.

### **AI Collaboration & Workflow**

I leveraged AI as a pair-programming partner to diagnose the scope error, pinpoint architectural SRP violations, and brainstorm potential refactoring strategies. Rather than accepting AI suggestions at face value, I independently evaluated the proposed updates and rigorously tested the final solution.

### **Manual Verification & Testing**

To confirm stability, I manually validated the implementation across standard, high-priority, and malformed task inputs. The queue processed valid items properly, rendered all expected log outputs, and initiated execution as intended. Additionally, I executed `node --check` to ensure the codebase remained free of syntax errors.